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Ελληνική Περίληψη

Η δερματοσκόπηση αντιπροσωπεύει μια μη επεμβατική μέθοδο που έχει κερδίσει σημαντικό ενδιαφέρον στον τομέα των φλεγμονωδών δερματολογικών παθήσεων. Αυτή η διπλωματική εργασία επιχειρεί να προσφέρει μια σύγχρονη και ολοκληρωμένη επισκόπηση της χρησιμότητας της δερματοσκόπησης στη διάγνωση της ψωρίασης. Ο στόχος της είναι να πραγματοποιήσει μια λεπτομερή ανασκόπηση της υφιστάμενης βιβλιογραφίας σχετικά με τη χρήση της δερματοσκόπησης στην αξιολόγηση της ψωρίασης. Η ερευνητική αξία της δερματοσκόπησης στην ψωρίαση κατά πλάκας, στην ψωρίαση των ονύχων και διάφορες άλλες εκδηλώσεις της ψωρίασης αναλύθηκε με την εξέταση αγγειακών μοτίβων, σχηματισμών λεπιών και πρόσθετων χαρακτηριστικών. Παράλληλα, η εργασία εξερευνάει την αξία της δερματοσκόπησης στη διαφορική διάγνωση, την αξιολόγηση της αποτελεσματικότητας των θεραπειών και της σοβαρότητας της κατάστασης, καθώς και την πρόγνωση και παρακολούθηση της ψωρίασης. Η δερματοσκόπηση έχει σημαντική κλινική σημασία στη διάγνωση και διαφορική διάγνωση της ψωρίασης και έχει δυνητικό επηρεασμό στην αξιολόγηση της σοβαρότητας και στην πρόβλεψη της αποτελεσματικότητας της θεραπείας τα τελευταία χρόνια.

Αγγλική Περίληψη

Dermoscopy represents a non-invasive modality that has garnered substantial interest within the realm of inflammatory dermatological conditions. This thesis endeavors to furnish a contemporary and comprehensive overview of the utility of dermoscopy in diagnosing psoriasis. The objective of this thesis is a meticulous review of extant literature about the employment of dermoscopy in the evaluation of psoriasis. The investigational merit of dermoscopy in psoriasis vulgaris, nail psoriasis, and various other manifestations of psoriasis was encapsulated by examining vascular patterns, scale formations, and additional features. Concurrently, the thesis is delving into the value of dermoscopy in differential diagnosis, assessment of treatment efficacy and severity, as well as prognostication and monitoring of psoriasis. Dermoscopy has demonstrated substantial clinical significance in the diagnosis and differential diagnosis of psoriasis and exhibits promising potential for severity assessment and efficacy prediction in recent years.

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1. INTRODUCTION

Psoriasis is a chronic, recurrent inflammatory skin condition mediated by the immune system, which can affect individuals across all age groups and impact the skin, nails, joints, and additional anatomical regions. Psoriasis affects approximately 2–4% of both males and females. It can manifest at any age, including childhood, with peak onset periods occurring between 15–25 years and 50–60 years. This condition tends to persist throughout an individual's life, varying in extent and severity. While psoriasis is particularly prevalent among Caucasians, it can affect individuals of any ethnicity. Approximately one-third of psoriasis patients have family members who also suffer from the condition. Immune factors and inflammatory cytokines, such as IL1 β , TNF α , IL-23, and IL-17, are responsible for the clinical manifestations of psoriasis. Consequently, these cytokines have become targets for biological drugs, leading to successful advancements in drug management. The hallmark clinical presentation of psoriasis is red, scaly plaques with well-defined edges and symmetrical distribution. The scales are typically silvery/white, the most common sites are the scalp, elbows, and knees, but any part of the skin can be affected. Itching is mainly mild but may be severe in some patients, resulting in scratching and lichenification characterized by thickened skin and skin markings. Painful skin fissures may occur, mostly on the palms and soles. Psoriasis can demonstrate the Koebner phenomenon. This phenomenon refers to the generation of new lesions on the skin that have been damaged/ irritated. Psoriatic plaques clear up, they may leave brown marks (post-inflammatory hypo- or hyperpigmentation) which can be expected to fade over several months. Additionally, the Auspitz sign refers to bleeding upon removal of the scaly layer in plaque psoriasis. It is related to the dilated capillaries of the dermis and is involved in the histological pathogenesis of chronic psoriasis.

Psoriasis is classified into acute (guttate) or chronic plaque type, early onset (<35 years of age – 75% of cases) or late onset psoriasis (> 50 years of age), localized or generalized, small plaque (<3cm) or large plaque psoriasis (>3cm), thin or thick plaques psoriasis and nail or no nail involvement. The types of psoriasis include guttate psoriasis, pustular psoriasis, plaque psoriasis

(psoriasis vulgaris), inverse (flexural) psoriasis, scalp psoriasis, nail psoriasis, and erythrodermic psoriasis. Psoriasis is usually associated with health conditions including psoriatic arthritis (autoimmune disease) and spondyloarthropathy (in 40% of patients with early-onset chronic plaque psoriasis), inflammatory bowel diseases, uveitis and celiac disease.

The diagnosis and evaluation of psoriasis are generally based on its characteristic clinical manifestations. In ambiguous cases, the diagnostic process may be supplemented by histopathological analysis, which, despite its utility, is an invasive procedure that many patients find unacceptable.

Dermoscopy is a noninvasive technique that not only magnifies the skin but also illuminates it, facilitating the visualization of the cornified layer and allowing for the examination of morphological details, such as pigmentary and vascular structures within the epidermis and superficial dermis, which are not discernible to the naked eye. Traditionally, dermoscopy has been employed in the diagnosis and assessment of malignant skin neoplasms; however, it has garnered increasing interest in the domain of inflammatory skin diseases, including psoriasis, in recent years. As a noninvasive ancillary diagnostic instrument, dermoscopy is generally more acceptable to patients and holds considerable promise for enhancing the diagnosis, evaluation, and management of psoriasis.

Despite the preliminary establishment of dermoscopic patterns associated with psoriasis, the clinical application of this technique for diagnosing and conducting differential diagnoses of the condition remains inadequately clarified. Additionally, there is a lack of comprehensive reviews addressing the significance of dermoscopy throughout the continuum of psoriasis. The objective of this thesis is to present a current overview of the role of dermoscopy in the diagnosis, differential diagnosis, efficacy evaluation, monitoring, and prognostication of psoriasis.

2. DIAGNOSIS

2.1 Psoriasis vulgaris

2.1.1 Vessel pattern

The systematic arrangement of red-dotted vessels represents the most prevalent vascular characteristic of psoriasis vulgaris, as observed through dermoscopy (Figure 1A, B). Blum et al. were the pioneers in identifying these symmetrically arranged, pinpoint-like capillaries via dermoscopy. This vascular pattern has frequently been referred to as dotted vessels, globular vessels, red dots, or red globules, with the corresponding histopathological features indicating the apical ends of vertically oriented dilated vessels that are uniformly distributed within the club-shaped dermal papillae. Numerous studies have documented the occurrence of red-dotted vessels in patients with psoriasis vulgaris at a frequency ranging from 96.0% to 100%, with approximately 55.6% to 90% of these vessels exhibiting regular distribution. Although dotted vessels constitute the most common vascular pattern found in psoriasis, they are not exclusive indicators of psoriasis vulgaris. Various studies have reported a diagnostic specificity of 74% associated with dotted vessels. Furthermore, these vessels have also been observed in certain inflammatory skin conditions, such as erythroplasia and lichen planus. Although the shape, size, and spacing of the vessels in these conditions exhibit heterogeneity, the specificity of the orderly distribution of vessels in the diagnosis of psoriasis has been determined to be 100%.

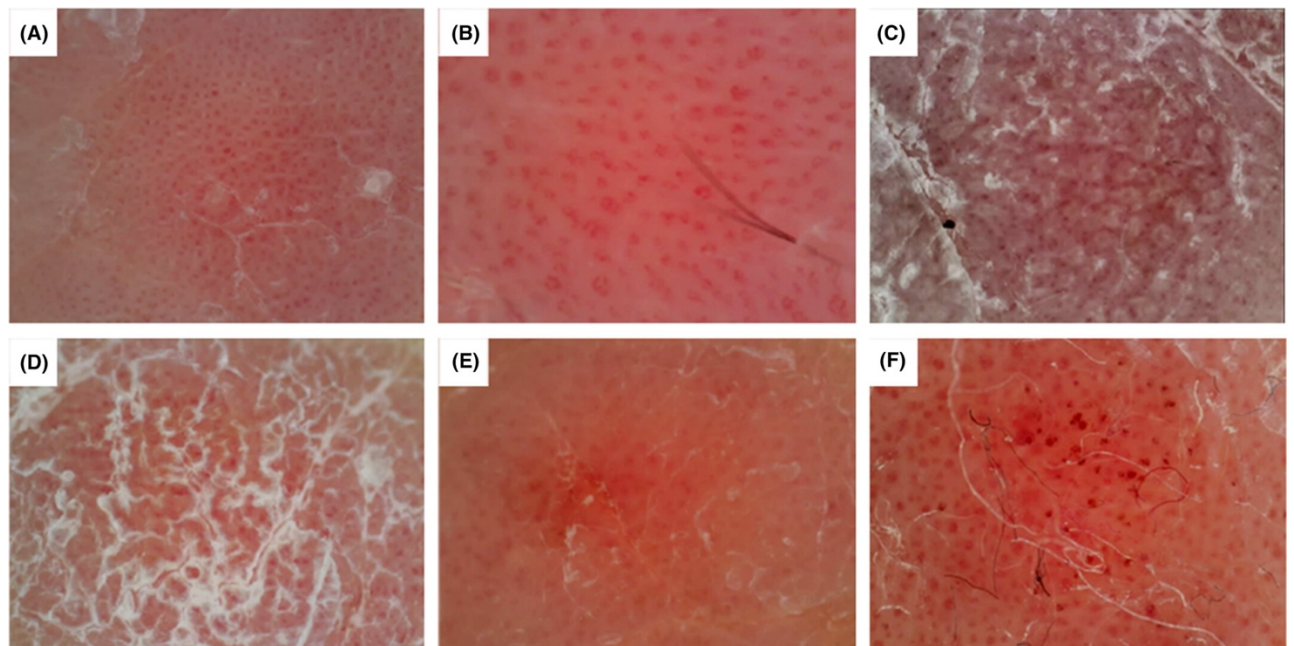


FIGURE 1

Dermoscopic manifestations of psoriasis vulgaris. (A, B) Regular arrangement of red-dotted vessels; (C) Ring vessels; (D) White scales; (E) Red background; (F) Hemorrhagic dots.

Differences in vascular patterns are noted based on the lesion's location and the skin phototypes. Within the palmar and plantar regions, the incidence of dotted vessels has been identified as the lowest, while in intertriginous areas (such as the axilla, submammary region, and groin), dotted vessels with regular distribution are more prevalent. In contrast, vessels located on the back tend to present as patchy. Observations of patients with darker skin types (V or VI) revealed patterns like those of individuals with lighter skin types (I-III); however, the frequency of specific vascular patterns was reduced. In a study conducted by Nwako-Mohamadi et al., red-dotted vessels were documented in only 64.2% of patients, regularly distributed vessels in 46.6%, while 35.8% exhibited no noticeable vascular changes.

Ring or hairpin-like vessels were initially documented by Vazquez-Lopez et al. Under dermoscopic examination, particularly at a magnification of 10 $\times$, these appeared as round capillaries (red globules) arranged in irregular circles or rings, exhibiting a beaded, lace-like capillary pattern (Figure 1C). Histologically, this vascular configuration represents tortuous, abundant, coiled, ectatic, and

elongated capillaries situated within the thin, elongated psoriatic dermal papillae, with variations in vessel shape resulting from the angle between the capillaries and the epidermis. In psoriasis vulgaris, the prevalence of ring/hairpin-like vessels is recorded at a rate of merely 3.6% to 44.1%; however, the diagnostic specificity of the ring-like vessels ranges from 94.6% to 100%, while that of hairpin-like vessels is 91.9%. Consequently, the identification of ring/hairpin-like vessels holds substantial significance in the diagnosis of psoriasis vulgaris.

2.1.2 Scale pattern

White scale is a characteristic manifestation of psoriasis (Figure 1D), with its corresponding histopathological features being orthokeratosis and/or parakeratosis. The diagnostic specificity of white scales for psoriasis vulgaris has been observed to be 83.8%. Under dermoscopy, white scales were identified in 64.7%–88.3%^[6-10, 12, 15] of patients with psoriasis. The scales were most commonly found on the scalp (85% to 100%)^[8, 9, 12] and the palmar and plantar regions (84.6% to 100%)^[8, 9], while scales on the face (54.4%), intertriginous areas (13.2%–77.8%)^[8, 9], and genitals (0.0%)^[8, 9] were relatively rare. The distribution of scales was predominantly diffuse (25.0%–70%)^[6-8, 12, 15, 16] and patchy (13.0%–55.0%)^[6-8, 12, 15]. Goliska et al. demonstrated that the scale distribution was mainly patchy (55.6%) in lesions lasting less than 5 weeks, whereas diffuse distribution was observed in 64.5% of older lesions.

In some patients with psoriasis, yellow scales were observed with or without white scales (2.0% to 25.2%)^[6, 7, 9, 10, 12, 15]. In young patients, yellow scales were more prevalent than in individuals over 50 years of age, potentially due to decreased skin turnover and increased hyperkeratosis associated with aging.

2.1.3 Other features

Red is the predominant background color for psoriasis (Figure 1E), and in pediatric and adolescent populations, some researchers have described the background color as milky pink. This background color has been shown to correspond to histopathological features such as thinning of the epidermis above the dermal papilla, dilatation of vessels in the dermal papilla and superficial dermis, and perivascular inflammatory cell infiltration. The sensitivity and specificity of the bright-red background were found to be 71.0% to 78.0% and 53.0% to 75.7%, respectively.

Hemorrhagic dots are dermoscopic manifestations of psoriasis (Figure 1F). Some researchers have also referred to them as the Auspitz sign or bleeding foci, which are significantly different from the dotted vessels typical of psoriasis. Bilgic et al. observed bleeding foci in 30% of patients with psoriasis, commonly in the lower legs and forearms, which may be related to blood extravasation caused by venous stasis and scratching.

In conclusion, the most common dermoscopic findings of psoriasis vulgaris are dotted vessels in a regular arrangement over a light red background and white scales in a diffuse arrangement. In a study by Lallas et al., if all three features were present, psoriasis vulgaris was highly predicted, with a diagnostic specificity of 88.0% and a sensitivity of 84.9%. Psoriasis vulgaris exhibits a specific dermoscopic examination pattern, with little heterogeneity between different lesion sites and skin phototypes. Dermoscopy is a tool of great clinical value in the diagnosis of psoriasis vulgaris.

2.2 Nail psoriasis

Dermoscopy is highly accurate in the diagnosis of nail psoriasis. Studies have demonstrated that in nail psoriasis, the concordance rate between clinical and dermoscopic examinations was 98%, and the concordance rate between histopathological and dermoscopic examinations was 88.88%. There were no significant differences in the agreement rates among clinical, dermoscopic, and histopathological examinations in diagnosing nail psoriasis.

Dermoscopy aids in identifying features that are not easily visible to the naked eye. Studies have shown that pitting, subungual hyperkeratosis, salmon patches, and splinter hemorrhages were better visualized using a dermoscopy than through clinical evaluation. Wannian et al. noted that the pseudofiber sign and dilated hyponychial capillaries were only visible with the aid of a dermoscope; consequently, the dermoscopically evaluated Nail Psoriasis Severity Index (NAPSI) scores were significantly higher than the clinically evaluated NAPSI scores. Additionally, Chauhan et al. observed that psoriatic features could also be detected in the clinically unaffected nails of some patients under dermoscopy, suggesting that dermoscopy may be beneficial in the early detection of nail psoriasis.

Dermoscopy is instrumental in determining the affected areas in nail psoriasis. Nail matrix involvement has been shown to manifest as deep pitting, red spots in the lunula, leukonychia, and similar features, while nail bed involvement manifests as salmon patches/oil drop signs, onycholysis, splinter hemorrhages, and other related features. Wannian et al. and Yorulmaz et al. observed filamentous red and black structures located near the cuticle or under the hyponychium along the distal free edge of the nail plate, which they termed pseudofibrous structures. They believed these to be bare capillaries, indicating nail bed involvement. However, Ankad et al. presented a contrasting perspective, asserting that these structures were adherent fibers rather than capillaries.

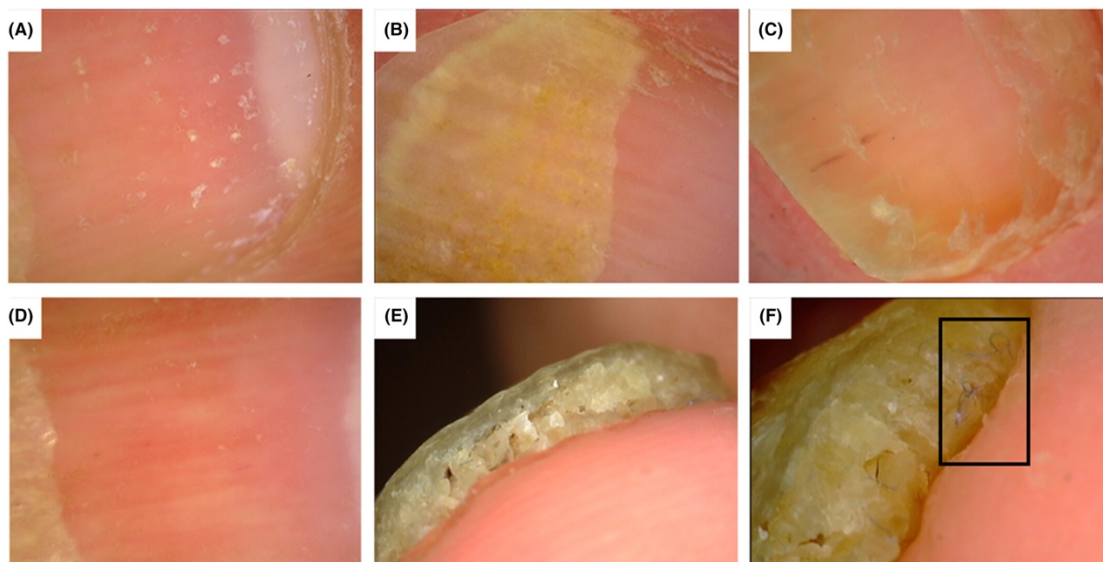


FIGURE 2

Dermoscopic manifestations of nail psoriasis. (A) Pitting; (B) Onycholysis; (C) Splinter hemorrhage; (D) Dilated hyponychial capillaries; (E) Subungual hyperkeratosis; (F) pseudofibrous structures (black box).

TABLE 1. Dermoscopic features and etiology of nail psoriasis.

Clinical manifestation	Involved areas	Dermoscopic features	Etiology
Pitting ^{23, 26, 28, 33}	Nail matrix	Large, irregularly distributed cupuliform depressions	Disruption in the keratinization process of the stratum corneum within the proximal nail matrix.

Red lunula ^{25, 31, 33}	Nail matrix	spotted or erythematous nail lunula	Intermediate and ventral matrix
Leukonychia ^{31, 33}	Nail matrix	Nail plates with white horizontal stripes	Parakeratoses of the distal nail matrix (leads to prevention of the normal desquamation of keratinocytes).
Beau's lines ^{31, 33}	Nail matrix	Transverse grooves and ridges within the nail plate	inflammation of the proximal nail fold.
Onychorrhexis ^{31, 33}	Nail matrix	Longitudinal splitting of nails	Corresponding longitudinal psoriatic lesions within the nail matrix.
Salmon patches/oil drop sign ^{26, 28, 33}	Nail bed	The yellow-red discolorations (visible as translucent and irregular areas on nail plate)	Parakeratosis within the nail bed.
Onycholysis ^{26, 28, 33}	Nail bed	Erythematous border encircling the white onycholytic area	Parakeratotic lesions of the hyponychium leading to separation of the nail plate from the nail bed allowing air to enter inside.

Splinter hemorrhage ^{23, 26, 33}	Nail bed	Reddish-brown or purplish-black striae arranged longitudinally (often on distal area of the nails)	Rupture of hyponychial capillaries leading to extravasation of blood down the grooves beneath the nail plate.
Dilated hyponychial capillaries ^{25, 27, 28}	Nail bed	The bright red dilated vessels are arranged parallel over the onychodermal band of the nail plate and surrounded by a prominent halo	Dilatation of nail bed capillaries
Subungual hyperkeratosis ^{25, 33}	Nail bed	The nail plate is elevated from the nail bed, displaying a white-silvery hue.	The deposition of cells under the nail plate which have not undergone desquamation.

2.3 Other types of psoriasis

Palmoplantar psoriasis manifests as hyperkeratotic, fissured plaques on the palms and soles, potentially impairing manual dexterity and ambulation. The predominant dermoscopic characteristic of palmoplantar psoriasis is diffuse white scaling (60.5%–100.0%), often associated with dryness and hyperkeratosis. Although red-dotted vessels (51.4%–90.9%) and a light-red background (48.6%–59.8%) are also prevalent, they are less commonly observed in these regions compared to other body sites, likely due to the thicker epidermis in the palmar and plantar areas, which obscures the visualization of underlying structures. Research by Yu et al. and Errichetti et al. indicates that the dotted vessels in palmoplantar psoriasis may be arranged in a beaded pattern along the sulci cutis. Although not always visible, this pattern has significant diagnostic specificity and is considered a crucial indicator for diagnosing palmoplantar psoriasis. (Figure [3A](#)).

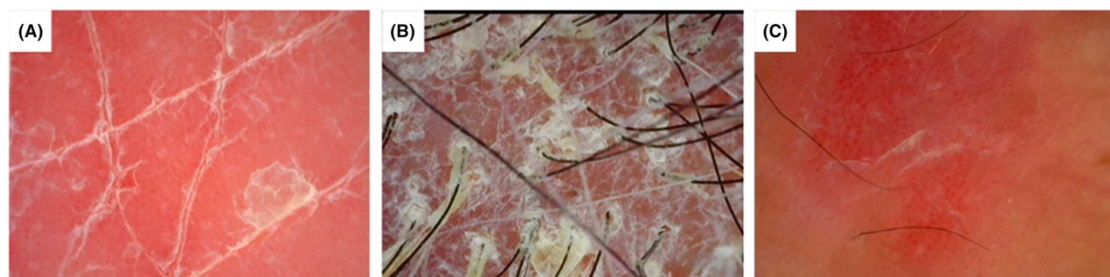


FIGURE 3

Dermoscopic manifestations of other type of psoriasis. (A) Palmoplantar psoriasis; (B) Scalp psoriasis; (C) Erythrodermic psoriasis.

The scalp is a prevalent site for psoriasis. The vascular pattern of scalp psoriasis closely resembles that of psoriasis vulgaris, with red dot/bulb vessels being the most common (94.0%). However, twisted red loops are more frequently observed in patients with scalp psoriasis (57.8%) compared to those with psoriasis vulgaris. Scales on the scalp are more commonly observed than

in other body regions, often presenting as white, diffuse or patchy, and thick. Additionally, Kibar et al. identified signet ring vessels and hidden hair as diagnostic features supporting the diagnosis of psoriasis.

Erythrodermic psoriasis is a rare and severe variant characterized by coalescent erythema, scales, or exfoliation involving at least 75% of the body. Although reports on the dermoscopic characteristics of erythrodermic psoriasis are scarce, only four cases have been documented, including two adult patients described by Errichetti et al. and Sławińska et al., and two 3-year-old Caucasian twins described by Campione et al. They noted that the dermoscopic pattern of erythrodermic psoriasis may align with that of psoriasis vulgaris, featuring regular, red-dotted vessels over a light-red background with white scales.

Approximately one-third of psoriasis patients develop psoriatic arthritis during their lifetime. Dermoscopy is also instrumental in diagnosing psoriatic arthritis. Zabotti et al. and Errichetti et al. highlighted that some patients with psoriatic arthritis exhibit features of nail psoriasis under dermoscopy. The presence of nail psoriasis has facilitated the early diagnosis of psoriatic arthritis, with a sensitivity and specificity of 96.0% and 83.3%, respectively. Furthermore, the use of hand ultrasound can increase specificity to 90.5%.

3. DIFFERENTIAL DIAGNOSIS

3.1 Differential diagnosis of psoriasis vulgaris

The observation of vascular morphology, vascular distribution, scale color, scale distribution, and background color with dermoscopy (refer to Table 2 and Figure 4) can serve as a foundation for the differential diagnosis of psoriasis vulgaris from lichen planus, pityriasis rosea, pityriasis rubra pilaris, and other dermatological conditions.

TABLE 3. Differential diagnosis of nail psoriasis.

	Dermoscopic features
Nail psoriasis ^{26-28, 33}	• Salmon patches • pitting

	• Brownish band between area of onycholysis and normal nail
Onychomycosis	• Spikes^{27, 56, 57}: in the proximal portion of the onycholytic space • Ruin pattern^{27, 58}: indentations on the ventral surface of the keratinocytes leading to accumulation of keratotic materials under nail plate • Longitudinal streak^{27, 56, 59, 60}: longitudinal streak on the nail plate, color ranging from bluish gray to black to whitish yellow
Traumatic onycholysis ^{27, 28}	Detachment of the nail plate from the nail bed
Allergic contact dermatitis due to artificial nails ^{62, 63}	• Onycholysis with a dented border • Periungual tissue destruction

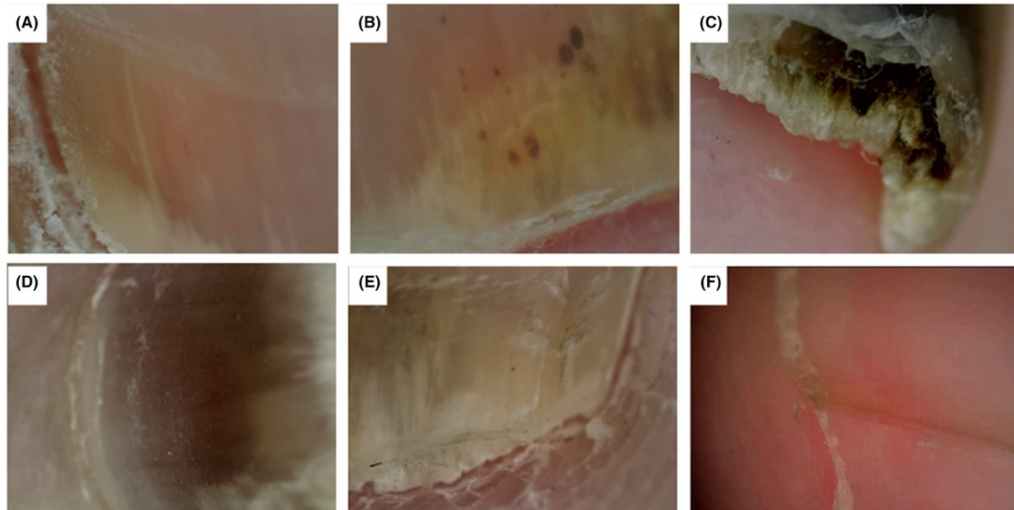


FIGURE 5

Differential diagnosis of nail psoriasis (A, B) Spikes in Onychomycosis; (C) Ruin pattern in Onychomycosis; (D) Longitudinal streak in Onychomycosis; (E) Distal irregular termination in Onychomycosis; (F) Traumatic onycholysis.

3.3 Differential diagnosis of palmoplantar psoriasis

The significant overlap in clinical features frequently complicates the differentiation between palmoplantar psoriasis and palmoplantar eczema (Figure 6A,C). Dermoscopy has emerged as an effective non-invasive method for differential diagnosis, potentially obviating the need for pathological biopsies. The predominant vascular morphology in both palmoplantar psoriasis and palmoplantar eczema is characterized by dotted vessels; however, a patchy distribution is commonly observed in palmoplantar eczema, contrasting with the regular distribution of vessels seen in psoriasis. Additionally, palmoplantar eczema is distinguished by a dull red or yellow background, brown or orange dots and globules, yellow crusts, and yellow-orange areas, which are infrequently present in psoriasis. Regarding scales, Errichetti et al. posited that the color of the scales serves as a crucial diagnostic clue. They asserted that yellow scales are prevalent in palmoplantar eczema, whereas white scales are typical in psoriasis. Nonetheless, Yu et al. observed that due to the local application of medications and other factors, the color of the scales

can sometimes be remarkably similar, thus rendering it an unreliable differentiator between palmoplantar psoriasis and palmoplantar eczema.

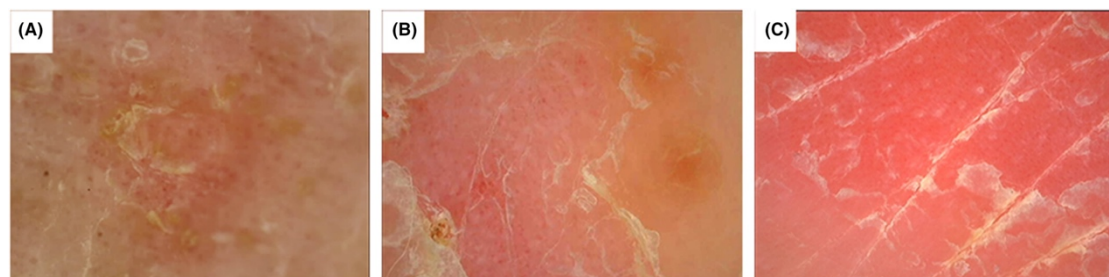


FIGURE 6

Differential diagnosis of palmoplantar psoriasis. (A) palmoplantar eczema; (B) eczema in psoriatico; (C) palmoplantar psoriasis.

Numerous studies have explored the concept of “eczema in psoriatico.” This term refers to patients with preexisting palmoplantar psoriasis who present with concurrent allergic contact dermatitis, exhibiting a type IV hypersensitivity pattern that perpetuates or triggers palmoplantar psoriasis. Overlapping dermoscopic features have been observed in eczema in psoriatico, including a regular distribution of dotted vessels and a diffuse distribution of white-yellow scales on a light red to yellowish dull red erythema background (Figure 6B). However, the body of research on eczema in psoriatico remains limited, necessitating further investigation and validation.

Dermoscopy also aids in distinguishing palmar syphiloderm from palmar psoriasis (Figure 6B). The background color serves as a crucial differentiating factor. The background of palmar syphilis is typically orange, attributed to the deposition of hemosiderin in the dermis, which contrasts markedly with the light-red background observed in psoriasis. Additionally, the presence of a circular, thin, scaling edge surrounded by an erythematous halo, known as the “Bielt sign,” has been identified as a significant diagnostic indicator of palmar syphiloderm. However, regularly distributed dotted vessels are observed in both palmar syphiloderm and palmoplantar psoriasis, rendering the vascular pattern an unreliable distinguishing feature.

3.4 Differential diagnosis of scalp psoriasis

Dermatitis is commonly observed on scalp, nasolabial folds, ears, eyebrows, and chest area. Although seborrheic dermatitis and psoriasis exhibit some distinct clinical differences, distinguishing between the two conditions when they are localized solely on the scalp can be challenging with the naked eye. Dermoscopy may prove beneficial in such cases. The vascular patterns observed in seborrheic dermatitis typically include arborizing red lines, comma vessels, and atypical vessels. Twisted red loops, which resemble the pattern observed in psoriasis, are relatively rare. While the incidence of white scales is comparable in both conditions, differentiation has primarily relied on vascular features. Utilizing the dermoscopic differential characteristics, a diagnostic method based on a deep learning model was developed to discriminate between scalp psoriasis and seborrheic dermatitis. This model demonstrated superior sensitivity, specificity, and area under the curve, outperforming the diagnoses made by five dermatologists with varying levels of experience.

4 SEVERITY AND THERAPEUTIC EFFECTS OF PSORIASIS

4.1 SEVERITY OF PSORIASIS

In terms of severity assessment, Carlesimo et al. proposed the Vascular Psoriasis Area Severity Index (VPASI) score, which is based on the Psoriasis Area Severity Index (PASI). This score incorporates vascular patterns and a more detailed partitioning of the body to provide a more sensitive and efficient quantification of psoriasis severity. However, Golińska et al. and Bilgic et al. noted that the PASI and Body Surface Area (BSA) were not correlated with any of the dermoscopic characteristics of psoriasis. Consequently, the utility of dermoscopy in assessing psoriasis severity requires further validation through additional studies.

Dermoscopy of the nail area is also valuable in evaluating psoriasis severity. Research has demonstrated that nail plate thickening and crumbling, trachyonychia, transverse grooves, salmon patches, and subungual hyperkeratosis are positively associated with the Nail Psoriasis Severity Index (NAPSI) score. Furthermore, several indicators, such as increased capillary density, red spots in the lunula, transverse grooves, nail plate crumbling, trachyonychia, and splinter hemorrhage, have shown a positive correlation with systemic inflammation. Multiple studies have reported a positive correlation between the PASI and BSA with the NAPSI, indicating that nail involvement is correlated with the severity of skin diseases. Therefore, the dermoscopic features of psoriatic nails are clinically significant, not only for assessing the severity of nail conditions but also for evaluating systemic inflammation.

4.2 THERAPEUTIC EFFECT OF PSORIASIS

Numerous studies have demonstrated that alterations in vascular patterns in psoriasis vulgaris can be utilized to evaluate treatment efficacy. Dynamic observations of patients with resolved psoriasis have shown that, with the amelioration of clinical symptoms, the diameter of the dilated glomus observed under dermoscopy gradually decreases. Furthermore, the distribution density of dotted/globular vessels diminishes progressively until there is a partial return to normal. The clinical symptom score has shown a significant correlation with the diameter of the glomus observed under dermoscopy, suggesting that changes in the vascular pattern in skin lesions may serve as an index to monitor the effectiveness of psoriasis treatment and the recovery of skin lesions. However, the color and distribution of scales are generally not employed in efficacy evaluations, likely due to the influence of scratching and clothing friction on the scale pattern. Researchers may sometimes select sites with fewer scales or remove scales to better observe the vascular pattern.

Although studies on the efficacy of nail dermoscopy are currently limited, they have shown promising clinical prospects. Iorizzo et al. reported that a significant reduction in the number of visible capillaries following the use of calcipotriol ointment in the treatment of nail psoriasis correlated positively with the degree of disease remission. Hashimoto et al. observed that after biological treatment

of nail psoriasis, symptom improvement was associated with diffuse scaling of the nail plate, transverse step-like notches, thickened white-yellow nail plate, elimination of splinter hemorrhages of the nail bed, and the development of erythematous borders of an onycholytic area. Khashaba et al. also found that in patients with nail psoriasis, dermoscopic features such as thickened white-yellow nail plates, scaling of the nail plate, and splinter hemorrhages had significantly improved after treatment with long-pulse Nd: YAG laser.

5. PREDICTION AND MONITORING OF PSORIASIS

5.1 PREDICTION OF THERAPEUTIC OUTCOME

In recent years, numerous studies have demonstrated a correlation between the treatment outcomes of psoriasis and specific dermoscopic features, such as hemorrhagic dots and blood vessel diameter. These dermoscopic features have emerged as potential predictors of the curative effects on psoriasis. Under dermoscopy, dotted vessels manifest as densely packed punctate structures. In contrast, globular vessels, a specialized subset of dotted vessels, present as twisted formations arranged in clusters reminiscent of glomeruli, with a diameter exceeding that of standard dotted vessels. Due to their similar appearance under handheld dermoscopy, earlier research did not deem the classification of dotted vessels by diameter to hold diagnostic significance. However, recent studies have shown that globular vessels tend to be associated with treatment resistance. In contrast, dotted vessels predict better treatment outcomes in local treatments, such as calcipotriene plus betamethasone dipropionate aerosol foam, narrowband ultraviolet B phototherapy, and 650-microsecond 1064-nm Nd:YAG laser. Several studies have found that globular vessels are more likely to be observed in the legs, where dotted vessels appear less frequently than in other parts of the body. This may be due to higher hydrostatic pressure resulting from microcirculation impairment in the legs, leading to increased vessel permeability and the consequent formation of a fibrin cuff around the capillaries, causing tissue hypoxia and the release of inflammatory mediators. However, the specific mechanism requires further study and verification.

Hemorrhagic dots are a common feature of psoriasis observed under dermoscopy. Lallas et al. found that, when using biological agents, the number and distribution of hemorrhagic dots may predict the treatment outcome 2–4 weeks in advance. The presence of hemorrhagic dots may serve as an early indicator of a favorable response to treatment, even preceding clinical improvement. This observation can reduce the likelihood of erroneously concluding that the drug is ineffective.

In the context of nail psoriasis, the study by Your et al.³⁰ indicated that a more favorable outcome might be anticipated when transverse grooves, longitudinal ridges, and discoloration are observed under dermoscopy. This suggests that dermoscopy could be instrumental in predicting the treatment outcomes for nail psoriasis.

Vazquez-Lopez et al.⁸³ observed that in patients with chronic psoriasis undergoing long-term topical steroid treatment, the appearance of red linear vessels under dermoscopy may predict imminent steroid-induced skin atrophy. This early detection can help identify lesions before they become clinically apparent, thereby preventing the progression of atrophy to a permanent state.

5.2 MONITORING AND PREDICTION OF RECURRENCE

In addition to efficacy predictors, specific dermoscopic features serve as monitoring factors for psoriasis and predictors of clinical recurrence. Several studies have demonstrated that the dermoscopic and clinical manifestations of psoriasis vulgaris and nail psoriasis can sometimes appear distinct. The recuperation of vessels or nails as observed under dermoscopy progresses more slowly than clinical recovery. Consequently, dotted vessels and partially healed surrounding skin may still be visible under dermoscopy, even when clinical skin lesions have fully resolved. The persistence of dotted vessels under dermoscopy post-treatment significantly increases the risk of disease recurrence compared to patients with complete vascular recovery. Consequently, some researchers have advocated for achieving “dermoscopic healing” rather than “clinical remission” in the treatment of psoriasis to ensure a more stable and prolonged curative effect. Furthermore, the recurrence of dotted vessels may predict clinical recurrence following recovery treatment with

biological agents, which could be instrumental in monitoring psoriasis post-recovery, early identification of disease recurrence, and guiding clinical treatment strategies.

6. FUTURE DIRECTIONS IN DERMOSCOPY FOR PSORIASIS

The field of dermoscopy is rapidly evolving, and its application in the management of psoriasis is no exception. As technology improves and our understanding of psoriasis deepens, dermoscopy is expected to play an even more crucial role in diagnosis, treatment monitoring, and personalized care. In this chapter, we will explore potential advancements and future directions in the use of dermoscopy for psoriasis.

Advancements in Dermoscopic Technology

i. Digital Dermoscopy and Artificial Intelligence

One of the most exciting developments in dermoscopy is the integration of artificial intelligence (AI) and machine learning (ML) with digital dermoscopic devices. AI algorithms can analyze dermoscopic images, identify specific patterns, and offer diagnostic suggestions that assist clinicians in identifying psoriasis early and accurately. These AI-based systems can process large datasets, learning from thousands of cases to recognize subtle features of psoriasis that might be missed by the human eye.

For psoriasis, AI systems may be able to:

- Detect early signs of psoriasis in atypical or small lesions.
- Automate the assessment of disease severity by recognizing changes in the dermoscopic features, such as scaling, erythema, and plaque thickness.
- Offer predictive analytics for monitoring disease progression and forecasting responses to treatment based on historical data.

ii. Enhanced Imaging Technologies

Recent advancements in dermoscopic imaging, such as **high definition (HD)** dermoscopy, allow for even more detailed visualization of skin lesions. Enhanced resolution can lead to better identification of the microvascular structures that are characteristic of psoriasis. With **polarized light** and **multispectral imaging**, dermatologists can capture a broader spectrum of skin surface patterns, allowing for a more nuanced understanding of psoriatic lesions. Additionally, **confocal microscopy** and **optical coherence tomography (OCT)** could enable dermatologists to visualize the psoriatic plaque at the cellular and tissue level in real time, providing greater insights into the inflammatory and hyperkeratotic changes occurring in psoriasis.

iii. Mobile	Dermoscopy	Devices
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As mobile technology continues to evolve, the use of smartphones and tablets in dermoscopy is expanding. **Mobile dermoscopy apps** now allow patients to capture high-quality images of their skin and share them with dermatologists remotely. This opens the door to telemedicine applications in psoriasis management, particularly for patients in remote areas or those with limited access to dermatological care. Mobile dermoscopy could allow for easier monitoring of psoriasis in routine follow-ups, increasing patient engagement and enhancing long-term disease management.

2. Personalized Psoriasis Management

*The future of psoriasis management is moving toward a **personalized medicine** approach, and dermoscopy can significantly contribute to this paradigm shift. By closely monitoring the changes in psoriatic plaques, dermoscopy can help dermatologists tailor treatment plans to individual patients.*

A. Biologic Therapies: *With the increasing use of biologics in psoriasis treatment, dermoscopy can help assess how well a patient is responding to these therapies. By tracking dermoscopic features such as plaque size, scaling, and vascular patterns, clinicians can adjust treatment regimens based on visual evidence of improvement or disease persistence.*

B. Topical vs. Systemic Treatments: *Dermoscopy can assist in determining whether a patient with psoriasis requires localized treatments (topical therapies) or systemic interventions based on lesion severity. It could provide objective data to support clinical decisions, especially in difficult-to-treat areas like the scalp and nails.*

3. Expanded Applications in Psoriasis Subtypes

a.	Scalp	Psoriasis
	Scalp psoriasis is often challenging to diagnose and manage due to hair coverage. Dermoscopy has already demonstrated utility in assessing scalp lesions, but future advancements in scalp-specific dermoscopy may provide more detailed visualization of the psoriatic plaques hidden beneath the hair. For instance, new dermoscopic tools might enable a clearer view of the crown vessels or glomerular patterns associated with scalp psoriasis, allowing dermatologists to monitor disease activity more effectively.	

b.	Nail	Psoriasis

Nail involvement in psoriasis presents another area where dermoscopy could evolve. While dermoscopy is currently used to identify signs of nail pitting, subungual hyperkeratosis, and oil-drop signs, further technological improvements may enable clinicians to assess the full extent of nail involvement in real time. **Nail dermoscopy** could help in differentiating between psoriasis and other nail conditions like fungal infections, which often present similarly.

c.	Pediatric	Psoriasis
1.	1. Diagnosis: Pediatric psoriasis is diagnosed based on clinical presentation, including well-demarcated, erythematous plaques with silvery scales, typically on the scalp, elbows, and knees. 2. Pathogenesis: Pediatric psoriasis is a chronic, immune-mediated condition. It is characterized by dysregulation of the immune system, leading to overproduction of skin cells and inflammation. 3. Management: Treatment options for pediatric psoriasis include topical corticosteroids, vitamin D analogs, and phototherapy. Systemic treatments like biologics (e.g., TNF inhibitors) may be used in severe cases.	1. Diagnosis: Psoriasis is diagnosed based on clinical presentation, including well-demarcated, erythematous plaques with silvery scales, typically on the scalp, elbows, and knees. 2. Pathogenesis: Psoriasis is a chronic, immune-mediated condition. It is characterized by dysregulation of the immune system, leading to overproduction of skin cells and inflammation. 3. Management: Treatment options for psoriasis include topical corticosteroids, vitamin D analogs, and phototherapy. Systemic treatments like biologics (e.g., TNF inhibitors) may be used in severe cases.

Pediatric psoriasis often presents atypically and is sometimes mistaken for other skin conditions. Dermoscopy, particularly when coupled with AI, could offer a powerful diagnostic tool to assist clinicians in differentiating pediatric psoriasis from other dermatological conditions, such as atopic dermatitis or fungal infections. Additionally, dermoscopy could play a role in monitoring disease progression in children, who may have a more dynamic disease course.

4. Research Opportunities and Clinical Trials

As dermoscopy continues to gain traction in psoriasis management, several areas remain ripe for further investigation:

a. Dermoscopic Markers of Treatment Response

Future research could focus on identifying specific dermoscopic markers that correlate with treatment response in psoriasis. For instance, studies could examine the relationship between the disappearance of specific vascular patterns or scaling under dermoscopy and clinical remission or disease flare. This could help dermatologists make more informed decisions about when to continue or adjust treatment regimens.

b. Multimodal Approaches to Psoriasis Monitoring

Integrating dermoscopy with other imaging techniques, such as **skin biopsy**, **ultrasound**, or **infrared thermography**, may allow for a more comprehensive assessment of psoriasis at various stages of progression. A

multimodal approach could offer deeper insights into the pathophysiology of psoriasis and provide better guidance for therapy selection.

c. Global Collaborations and Data Sharing

In order to improve the accuracy and reproducibility of dermoscopy in psoriasis diagnosis, global collaboration in terms of data collection and image sharing will be crucial. Creating large, publicly available datasets of dermoscopic images of psoriasis will allow for the development of more robust AI algorithms and better understanding of global patterns of psoriasis presentation, especially in underrepresented populations.

7. CONCLUSION

Dermoscopy has emerged as a valuable tool in the diagnosis, monitoring, and management of psoriasis. Over the past two decades, significant advancements in dermoscopy technology, combined with a deeper understanding of the dermoscopic patterns associated with psoriasis, have allowed dermatologists to enhance the accuracy and efficiency of their clinical practice. This study has demonstrated that dermoscopy provides key insights into the early diagnosis of psoriasis, particularly in its more subtle forms or atypical presentations, where clinical differentiation from other dermatological conditions can be challenging. Dermoscopy facilitates the identification of hallmark psoriatic features such as the characteristic silvery scales, dotted vessels, and the presence of crown vessels, all of which play a critical role in confirming the diagnosis. The ability to visualize microvascular changes in psoriasis lesions using dermoscopy enables clinicians to assess disease severity and monitor disease progression in a non-invasive and real-time manner. The dynamic nature of psoriasis, with its varied clinical presentations and response to treatment, demands precise tools for continuous evaluation, and dermoscopy fits this need effectively. Moreover, dermoscopy's utility extends beyond diagnosis, offering a powerful tool for tracking treatment responses. In particular, digital dermoscopy, which allows for the comparison of sequential images, offers significant advantages in evaluating the effects of

systemic treatments, biologic therapies, and topical regimens on psoriatic plaques. The ability to observe changes in lesion characteristics—such as reduced vascularity, thinning of scales, or the disappearance of erythema—offers valuable objective data that complements clinical assessments. Such objective monitoring can guide clinical decisions regarding the continuation or modification of treatment plans, optimizing therapeutic outcomes for patients. However, while dermoscopy has proven invaluable in these contexts, it is important to acknowledge several limitations. Despite its high sensitivity and specificity, dermoscopy is not without challenges, particularly in cases of extensive or overlapping skin conditions. Moreover, the need for specialized equipment and trained professionals to perform dermoscopy is a notable barrier to widespread adoption in non-specialized settings. The interpretation of dermoscopic findings requires substantial expertise, and there is a need for standardization of dermoscopic features and scoring systems to ensure consistency and reliability in clinical practice. Looking forward, the integration of artificial intelligence (AI) into dermoscopy is poised to revolutionize psoriasis management. AI-powered dermoscopic analysis could potentially enhance diagnostic accuracy, enabling automated detection of subtle dermoscopic patterns that may be overlooked by human examiners. Furthermore, the use of AI to monitor disease progression could facilitate more personalized treatment approaches by identifying patients who may benefit from more aggressive or tailored therapeutic regimens. The convergence of AI, machine learning, and dermoscopy in psoriasis care represents an exciting frontier that could significantly improve patient outcomes, particularly in the context of precision medicine. Additionally, the advent of teledermatology and mobile dermoscopy has opened new possibilities for managing psoriasis in remote or underserved areas. The ability to capture high-quality dermoscopic images and transmit them for remote evaluation by dermatologists is an important step toward making psoriasis care more accessible to a global population, especially in rural or developing regions. This approach also enhances patient engagement by enabling more frequent monitoring of lesions without the need for regular in-person consultations. Dermoscopy has firmly established itself as a vital tool in the armamentarium of dermatologists treating psoriasis. Its ability to offer a detailed, non-invasive assessment of psoriatic lesions provides clinicians with

critical information for both diagnostic and therapeutic purposes. While challenges in its widespread adoption remain, especially in terms of accessibility and expertise, the ongoing advancements in dermoscopic technology, combined with the integration of AI and teledermatology, promise to further enhance the precision and personalization of psoriasis care in the near future. By continuing to refine its application, dermoscopy will undoubtedly play an even greater role in the evolving landscape of psoriasis management.

8. CONFLICT OF INTEREST STATEMENT

No conflicts of interest declared.

9. ETHICS STATEMENT

This diploma thesis is based on previously conducted research and does not include any studies involving human participants or animals performed by the author.

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DECLARATION

I, Maria Nektaria Giannakaki the undersigned, hereby declare that I have independently completed the diploma work. I am aware of the potential consequences that may arise in the event of a breach of this declaration.